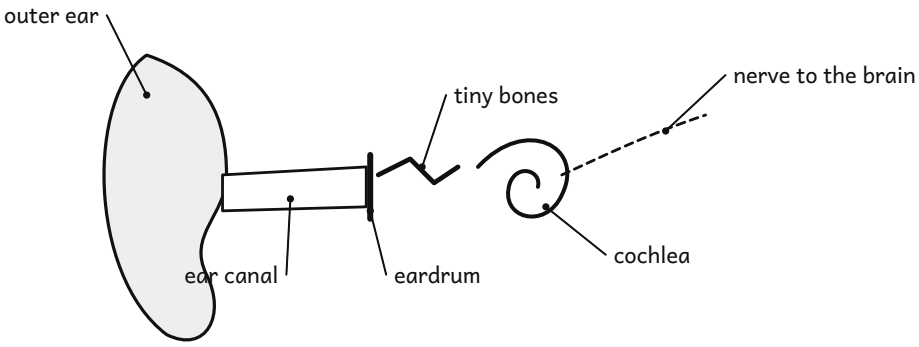
PHYSICAL AND CHEMICAL SCIENCE

Hearing and Sound

The parts of the ear, why hearing matters, noise pollution, and sources of light and sound



In this pack

- **Learn it:** How we hear ★☆☆
- **Fact check:** Parts of the ear ★☆☆
- **Think it through:** Listening and staying safe ★★☆
- **Investigate:** Which material blocks sound best? ★★☆
- **Real-world case study:** Quiet near the hospital ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science: Stage 3 Unit 4 Our five senses (hearing).

This pack belongs to: _____ **Class:** _____

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LEARN IT · C3 · HEARING AND SOUND

How we hear

★☆☆
Grade 2

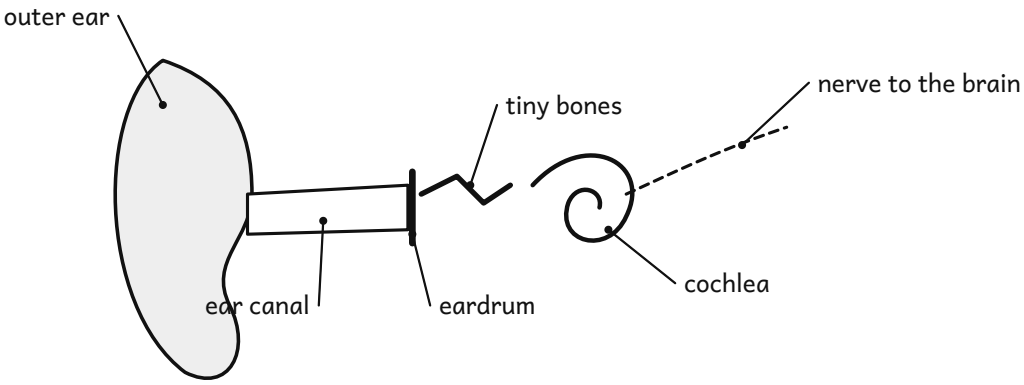
Name: Class: Date:

Learning goals

- Name the parts of the ear and explain how we hear.
- Explain why hearing is important for people and animals.
- Talk about noise pollution and ways to block sound, and name sources of light and sound.

Ring a bell, beat a dhol, sing a song. All sounds are made when something **vibrates** (shakes very quickly). The sound travels through the air to your **ears**.

Inside the ear



The parts of the ear.

- The **outer ear** collects sound and sends it down the **ear canal**.
- The sound makes the **eardrum** vibrate, like a tiny drum. This is in the **middle ear**, with three **tiny bones**.
- In the **inner ear**, the snail-shaped **cochlea** turns the vibrations into messages.
- A **nerve** carries the messages to the **brain**, and the brain tells you what you hear.

Why hearing matters

Hearing helps us **talk** and **listen** to each other, enjoy **music**, and stay **safe**: we hear a car horn, a fire alarm or someone calling. Animals use hearing too: a deer listens for a tiger, an owl hunts mice by listening in the dark, and bats find insects using echoes.



listens for danger



hunts by hearing



uses echoes



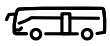
hears very quiet sounds

Too loud!

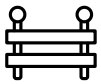
Very loud sound for a long time can **damage** your hearing. Unwanted loud sound is called **noise pollution**: horns, loudspeakers, machines and generators. We can block sound with **soft** materials like curtains, foam and ear muffs, by closing windows, or by moving further away.



loudspeakers



horns



machines



ear muffs block sound

Sources of sound and light

A **source** is where something comes from. **Sound sources** include drums, bells, birds and voices. **Light sources** include the Sun, lamps, torches and candles.

⚠ Stay safe

Never put anything inside your ear. If a sound is so loud it hurts, cover your ears and move away.

Key words

vibrate	shake backwards and forwards very fast
eardrum	thin skin in the ear that vibrates
cochlea	snail-shaped part of the inner ear
noise pollution	too much unwanted loud sound
source	where light or sound comes from

FACT CHECK · C3 · HEARING AND SOUND

Parts of the ear

★☆☆ Grade 2

Name: Class: Date:

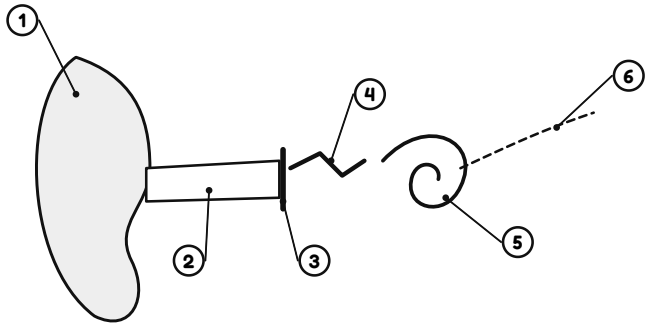
Learning goals

- Label the ear.
- Sort sources of light and sound.

1 Label the ear. [6]

LABELS

eardrum **nerve to the brain** **ear canal** **tiny bones** **outer ear** **cochlea**



1 _____ 2 _____ 3 _____

4 _____ 5 _____ 6 _____

2 Sort the sources. [6]

SORT THESE

bell **drum** **Sun** **candle** **bird** **lamp**

Sound source	Light source

3 Fill in the gaps. [3]

WORD BANK

vibrate **eardrum** **nose** **nerve**

Sounds are made when things _____. Sound makes the _____ vibrate. A _____ carries messages to the brain.

THINK IT THROUGH · C3 · HEARING AND SOUND

Listening and staying safe

★★☆
Grade 2

Name: Class: Date:

Learning goals

- Explain how hearing helps.
- Suggest ways to block noise.

1 **How does hearing keep you safe when you cross the road?** [2]

2 **Your classroom is next to a noisy road. Give two ways to make it quieter.** [2]

3 **Why do workers using loud machines wear ear muffs?** [2]

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LEARN · PLAY · WONDER

Which material blocks sound best?

★★☆
Grade 2

Name: **Class:** **Date:**

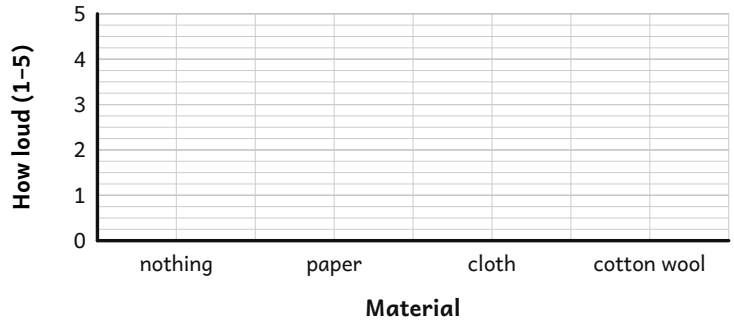
Learning goals

- Do a fair test.
- Compare results.

Class 2C put a ticking clock in a box and covered it with different materials. A child stood at the same place and said how well they could hear it.

1 **Complete the table and draw a bar chart (5 = very loud, 1 = very quiet).** [3]

Material	nothing	paper	cloth	cotton wool
How loud (1–5)	5	4	3	1



2 **Which material blocked the sound best?** [1]

.....

3 **What did they keep the same?** [1]

.....

REAL-WORLD CASE STUDY · C3 · HEARING AND SOUND

Quiet near the hospital

★★★ Grade 2

Name: **Class:** **Date:**

Learning goals

- Think about noise and health.

Science | Health | Citizenship

Near a hospital in Dhaka there is a sign: **"Silent zone. Do not use the horn."** Sick people need quiet so they can rest and get better.

But many drivers still honk their horns. Children from a nearby school made posters and talked to drivers. They asked them to keep the area quiet. Some drivers stopped honking.

A story about noise near a hospital.

1 **Why do people in hospital need quiet?** [2]

2 **What did the children do?** [2]

3 **Draw a poster to ask drivers to keep quiet.** [2]

My poster

Show what you know

★★★
Grade 2

Name: Class: Date:

Learning goals

- Use sound words.

1 Find the words.

S	H	N	O	I	S	E	A
O	B	G	L	V	T	I	F
U	A	Q	E	E	J	U	P
N	T	U	O	Y	A	M	C
D	S	I	D	N	Y	R	L
B	D	E	R	F	M	W	O
H	L	T	U	C	R	D	U
J	K	M	M	G	I	C	D

drum ear loud noise quiet sound

2 Which part of the ear vibrates like a drum? [1]

- ◯ eardrum
- ◯ nose
- ◯ tongue

3 Which is best for blocking noise? [1]

- ◯ glass
- ◯ thick cloth
- ◯ metal

4 Challenge. Draw three things that make sounds in your home. [3]

Sounds at home

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
label the parts of the ear	○	○	○
explain how we hear	○	○	○
say why hearing matters	○	○	○
name ways to block noise	○	○	○
sort sources of light and sound	○	○	○